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Feasibility assessment of the use of steam sourced from nuclear plants for French factories considering spatial configuration

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ABSTRACT

This paper provides the basis for the analysis and planning of industrial symbiosis in France, through the most efficient use of existing technologies. The French power mix has largely relied on nuclear plants and, despite an expected decrease of the installed capacity, these thermal plants could remain in the long term. Nuclear plants emit a significant amount of heat into the surrounding environment. Transferring some of this heat to nearby industrial sinks could lead to a reduction in fossil fuel consumption and CO_2 emissions. Relevant industrial subsectors for using 250 °C power plant-sourced steam are determined based on empirical data at a 5-digit level of disaggregation of NACE revision 2 (including e.g. pulp & paper and basic organic chemicals). The regional and sectoral distribution of relevant facilities is then analysed using spatial mapping methods. Lastly, a techno-economic model is built to assess the feasibility of transferring steam sourced from nuclear plants to industrial users. Compare to on-site steam generation, power plant-sourced steam supply could reduce energy supply costs and CO_2 emissions in locations close to these plants. Optimum relocation of factories could make steam exchanges more cost-effective, and such considerations should be incorporated in future local planning strategies.

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1. Introduction

1.1. Background

There is growing concern about the unsustainability of the economic model in developed countries, in particular with regard to climate change (EC (European Commission), 2014a; Lockie and Sonnenfeld, 2013). The literature on industrial ecology emphasises that improved energy efficiency and implementation of low fossil-carbon energy based systems is the most direct and effective way of reducing equivalent-CO_2 emissions in the industrial sector (Chang, 2015; Chertow and Lombardi, 2005; Huisingh et al., 2015). In France, the electricity has a relatively low average CO_2 content compared to other European countries (62 t eCO_2/GWh_{th}, while the EU average is 347 t eCO_2/GWh_{th} (Ministère de l'environnement, de l'énergie et de la mer (French Ministry of Environment, Energy and Seas), 2017). At the same time, the energy consumption within the industrial sector is responsible for 13.1% of total equivalent – CO_2emissions (Ministère de l'environnement, de l'énergie et de la mer, 2017). The fundamental idea of this paper is therefore to explore the opportunity for decarbonising the industrial sector through efficient use of existing technologies. This could include, in France, the use of heat sourced from nuclear plants for industrial applications.

Towards 2025–30, the nuclear installed capacity should remain significant (40–63 MW_e). This despite plans to reduce the share of nuclear production in the electricity mix from 75% (2016 level) to 50% (2025), as targeted by public authorities (Ministère de l'environnement, de l'énergie et de la mer, 2015). In the longer term, dispatchable power plants may present some new benefit to balance power systems with large amounts of intermittent renewables (Cany et al., 2016; Wang et al., 2016). Given the path dependence approach (past investments made), as well as the low carbon profile of this energy source, nuclear plants could remain in use for a number of decades in France (see e.g. Maïzi and Assoumou, 2014). In order to enhance the development of industrial ecology

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practices, some of these plants could be optimised to supply heat to nearby industrial sites.

Nuclear plants can be designed to provide both electricity and heat without jeopardising nuclear safety (STUK (Radiation and Nuclear Safety Authority of Finland), 2009: p. 6). Operating thermal plants in cogeneration mode enables the recovery of some of the thermal energy which is otherwise wasted in the surrounding environment. While the nuclear cogeneration application in which there is most experience is district heating (Leurent et al., 2017, 2018a; 2018b), there is also some experience of steam supply to nearby industrial facilities, as shown in Table 1. The French nuclear fleet, and most nuclear reactors operating in the world today (277 out of 438) and under construction (59 out of 69; IAEA, 2017) are Pressurised Water Reactors (PWR). In PWR, the steam at the steam generator outlet is superheated to 285 °C. In France, the consumption of heat below 250 °C represents approximately one third of the final energy used in the industrial sector (excluding energy production and refineries) and the bulk of the demand ranges from 80 °C to 250 °C (ADEME (French Environment and Energy Management Agency), 2015). Hence, it is clear that the feasibility of 250 °C steam transfer from nuclear plants to industrial facilities is worth assessing for France. This paper is dedicated to this objective.

1.2. Research scope

Industrial symbiosis (IS) is one of the key concepts for using energy and resources effectively. Chertow (2000) defined IS as engaging "traditionally separate industries in a collective approach to competitive advantage involving physical exchanges of materials, energy, water, and/or by-products." Reported examples of IS include energy exchange in chemical clusters (Shi et al., 2010), iron/steel industrial parks (Zhang et al., 2013), between power plants and industrial/residential areas (Jacobsen, 2006; Chertow and Lombardi, 2005). In France, there are currently 14 IS complexes and more than 60 new cooperations are being considered (OREE (French association of companies and local municipalities promoting circular economy), 2015). Although French complexes have focused mainly on effective material use, there is an increasing policy expectation that exhaust heat from power plants and industries should be valorised (Ministère de l'environnement, de l'énergie et de la mer, 2014: p. 14). In the town of Dunkirk, France, waste heat from a steel manufacturing plant is distributed to the city district heating network. This practice has reduced the energy demand and CO_2 emissions by 128 GWh_{th}/year and 30 kt eCO_2/year respectively, compared to the use of conventional energy systems (CEREMA (French Research Centre on Risks, Environment, Mobility and Territorial Planning), 2009).

However, heat exchange synergies are still rare in France. IS projects are complex systems that may have difficulty breaking through established routines and business models (Tudor et al., 2007). The literature on IS emphasises the need to identify key organisations in a region around which the recycling network of industrial players could emerge (Wallner, 1999; Boons and Baas, 1997). According to Korhonen (2001), a combined heat and power plant can serve as a starting point for developing an IS strategy. According to these results, it seems that the implementation of IS in France could be facilitated by the use of steam sourced from nuclear plants. Nuclear plants could serve as the driver of some of the main material and energy flows of the regional energy supply system, and as a possible organisation around which the control and management of these flows are organised. This may be made easier by the fact that the French nuclear operator is a public company.

Yet, such projects can face considerable obstacles. Despite being widely available, there is no actual experience of nuclear plant-sourced heat utilisation in France. The attractiveness of this alternative is lessened by the policy challenges inherent in its wide diversity (Leurent et al., 2017). From a systemic point of view, the main investment involved in this kind of project will be in the pipeline required to transport the steam from the heat sources to the sinks. Unlike electricity, which has a well-developed grid network, new pipeline infrastructures must be carefully planned. It may not be cost-effective to transport the steam over distances of more than 3–5 km (NISP (National Industrial Symbiosis Project), 2008). The proximity of energy consumers to suppliers is therefore a key design factor for IS systems. To establish sustainable IS complexes in a low-carbon scenario, it is necessary to identify factories near power plants that may operate until 2050, or to establish power plants near the locations of future IS complexes. Since new nuclear plants, if they are commissioned, will probably replace decommissioned plants on existing sites, the former case is more likely.

Therefore, it is necessary to combine spatial planning and energy system modelling methods. Some studies have discussed the feasibility of using heat from power plants and have demonstrated considerable energy potential for each analytical area (Seck et al., 2015; Bowman, 2012; Safa, 2012). Wang et al. (2017) proposes a model to improve the simulation of steam transportation pipelines that are crucial for heat exchanges in IS complexes. However, most of the previous research focused on the optimisation of existing heat exchanges with static systems, and only a few studies have considered land use planning issues. Fixed demand is not a reasonable assumption when assessing the long-term dynamics of the energy system, and the consideration of relocation issues is important. In this respect, the study led by Togawa et al. (2014) is significant. Using spatial mapping methods, Togawa et al. (2014) showed that the relocation of a tomato factory to within a 2 km radius of a thermal plant could reduce energy costs while reducing equivalent-CO_2 emissions by 47% compared to the use of an individual fuel oil-based boiler.

Table 1
Experience of industrial use of steam sourced from nuclear plants. Data source: Verfondern (2013).

Country/reactor type	Net power output (MWh_e)	Heat delivery (MWh_{th})	Supply temperature (°C)	Distance to factory (km)	Industrial application	Operation period of application
Norway/Halden BWR ([a])		1 × 20	–	Adjacent site	Paper mill	1964 -
Switzerland/Gösgen PWR	1 × 985	1 × 45	220	1.8	Cardboard factory	1979 -
Canada/Bruce CANDU ([b])	4 × 860	4 × 848	–	On-site	Heavy water and others	1981–1997
Germany/Stade PWR	1 × 630	1 × 1892	190	1.5	Salt refinery	1984–2003

Notes:
[a] BWR: Boiling Water Reactor.
[b] CANDU: CANada Deuterium Uranium.

From an academic perspective, there is a need to further investigate the techno-economic feasibility of potentially new IS systems so as to inform policy makers and stakeholders on the cost and climate savings potential of such systems. However, accurate modelling of IS complexes requires the collection of a substantial amount of data. Information such as the size of the factories and the location of the plants in relation to one another would provide information on ease of access. In addition, the viability of the project cannot be established unless industries provide details on planned and unplanned plant outages. The information on source availability is therefore crucial for designing IS complexes but can be difficult to collect due to confidentiality issues. In that respect, a key added value of our research is the accuracy of the data used, which covers all French factories at a 5 digit-level of disaggregation of NACE revision 2, with 114 subsectors, for 2012.

1.3. Outline of the article

Based on these research statements, this study aims to fill the research gap in three areas. Firstly, relevant subsectors for the use of power plant-sourced steam are determined. Secondly, combined with spatial analysis, the geographical and sectoral distribution of relevant factories is assessed. Thirdly, cost-effective land use considering relocation of the energy demand is analysed using two applied cases, so as to reduce equivalent-CO_2 emissions through the relocation of high-energy industrial consumers closest to the heat sources. Section 2 describes our key data, the data processing method and additional modelling assumptions. Then Section 3 gives the results and discusses them. Section 4 concludes.

2. Data processing and modelling assumptions

The methodology followed can be broken down into three consecutive steps. Step 1 uses our key data to characterise those subsectors which constitute relevant targets for replacement of conventional, individual steam generation systems with external steam supply. Following geo-referencing of all production facilities and nuclear sites in France, step 2 determines the geographical and sectoral distribution of the factories in the subsectors identified in step 1. Lastly, step 3 builds a model designed to assess the techno-economic potential of IS complexes using power plant-sourced steam.

2.1. Identification of the relevant industrial subsectors for external supply of 250 °C steam

To build a prospective scenario for decarbonising the industrial sector, it is essential to have detailed information on the CO_2 emission factor of conventional heating systems, the frequency of factory shutdowns and the range of temperatures required. It is therefore important to understand the data used in this paper and the processing method used.

2.1.1. Presentation of the data

The core data used is based on a comprehensive study initiated by the French National Alliance for Energy Research Coordination (ANCRE , 2013). The database is constructed from more than 6000 data points incorporating energy knowledge of industrial sectors from the National Centre for Analysis and Research on Energy (CEREN) in 2012. These data were provided at a 5-digit level of disaggregation of the NACE revision 2 classification with 114 subsectors (EC , 2008). The information for each subsector concerns the amount of heat (GW_{th}/year) that was:

- Generated using a particular type of energy: steam or other heating medium, natural gas, electricity, renewable energies, oil products or a mixture of these sources. Given that the direct and lifecycle CO_2 emission factors of these energies are known (see Table 2), the average equivalent CO_2 emission factors of heat generation in factories can be assessed for each subsector;
- Consumed within a particular temperature range: 70 °C, 100–150 °C, 150–250 °C, or >250 °C;
- Used to operate a factory with a given working schedule: 1 × 8 h, 2 × 8 h, 3 × 8 h with stops, 3 × 8 h without stops. The average number of hours during which the factories in each subsector were operated was also available (<2000 h, 2000–4000 h, 4000–7000 h, >7000 h). This data enables to calculate $NH_{FT,j}$, the average number of hours per year during which factories in subsector j operated.

2.1.2. Data processing using economic criteria

The higher the energy intensity, the higher the competitiveness risk related to CO_2 pricing policies (Meleo, 2014), and the greater the benefits expected from energy cost reduction. In order to provide insights into the potential interest of factory owners in reducing energy costs, EI_j, the average energy intensity observed in subsector j, is calculated using equation (1):

$$EI_j = \frac{\sum_e G_{e,j} P_e}{PV_j} \quad (1)$$

Where $G_{e,j}$ is the total amount of heat generated using energy e in subsector j ($GW_{th}/year$); P_e is the price of energy e observed in 2012; and PV_j is the total production value of subsector j in 2012 (available from INSEE (French National Institution for Statistics and Economics Studies), 2016). Following the taxonomy established by Seck et al. (2015), this paper studies both energy intensive ($EI_j \geq 5\%$) and non-energy intensive subsectors. We have however discarded the subsectors which would show little interest in reducing energy costs ($EI_j < 1\%$).

We have chosen to use the production value as the preferred indicator of economic activity for the rest of the study. In our case, using the production value is preferable due to the difficulty of comparing added values across diverse subsectors (Seck et al., 2015).

2.1.3. Data processing using environmental criteria

External steam supply is not the only way of improving the industrial energy efficiency. Factory owners should also consider internal energy efficiency investments, through the implementation of the technologies listed by e.g. Hasanbeigi et al. (2016) or Kwak et al. (2014). Despite the potential benefits, external steam supply may inhibit the will of factory owners to invest in internal energy efficiency technologies because they have invested in capital-intensive systems (e.g. steam transportation system), and are waiting for the payback from their investments (Korhonen, 2001). Following discussions with experts from CEREN and ANCRE, we have discarded factories that primarily consume heat above 250 °C (>50% of total consumption).

2.2. Determining the spatial and sectoral distribution of relevant factories

While step 1 considers industries at a 5-digit level of NACE Rev. 2, step 2 goes down to factory level. The CO_2 emission factors, working schedule and ratio of heat costs over production value for specific factories are assumed to be equal to the average for the subsector (see 2.1). However, additional processing was performed to establish the heat consumption within all French industrial facilities. Firstly, C_j, the average heat consumption of a factory in subsector j (GW_{th}/year)is expressed in equation (2):

Table 2
Direct and lifecycle CO_2 emission factors of energy sources. Data source: IPCC (Intergovernmental Panel on Climate Change, 2012); EEA (European Environment Agency), 2011.

Energy source	Equivalent-CO_2 emission factor ($t\ eCO_2/GWh_{th}$)
Natural gas	469
Oil products	750
Heating fluid (mostly oil products, then gas)[a]	600
Mixed (mostly gas, then oil products)*	500
Renewable heat sources	15
Electricity (average in France)	150

Notes:
[a] Discussions with experts from CEREN and ANCRE helped with the formulation of these assumptions.

$$C_j = \frac{TC_j}{N_j} \tag{2}$$

where, TC_j is the total heat consumption in subsector j (GW_{th}/year); and N_j is the total number of facilities in subsector j (INSEE, 2016).

Then, $C_{i,j}$, the heat consumed by factory i in subsector j (GW_{th}/year), is expressed in equation (3):

$$C_{i,j} = C_j \frac{W_{i,j}}{W_j} \tag{3}$$

where C_j, is the average heat consumption of a factory in subsector j ($GW_{th}/year$); $W_{i,j}$ is the workforce of factory i in subsector j (INSEE, 2016); and W_j is the average workforce of a factory in subsector j (INSEE, 2016). The assumption supporting equation (3) is that the factories in each subsector are relatively homogeneous in terms of the amount of heat consumed per employee. However, in a new factory one can expect technical progress to reduce the workforce required to produce a fixed output. In order to minimise this potential effect, this paper uses the more detailed disaggregation level of NACE rev. 2 (5-digit level) for the French industrial sector.

Using spatial mapping methods, the datasets produced for each French factory were plotted on a boundary map of France using a Geographical Information System. Together with the georeferencing of nuclear sites, the methodological use of our data enables us to assess the geographical and sectoral distribution of all facilities that are included in a subsector identified in step 1. The assessment boundaries are restricted to factories that are:

- Consuming more than 10 GW_{th}/year of heat below 250 °C. In practice, IS systems are mostly composed of large energy consumers (Chertow, 2000) as there may be a critical size for a company to support the related costs and be involved in long-term agreements with other firms (Maes et al., 2011);
- Located less than 100 km from a nuclear site. When several sites were located within a 100 km radius of the company the closest site was chosen.

2.3. Assessing the techno-economic feasibility of transferring power plant-sourced steam to factories

In step 3, an energy model based on process analysis is established to assess an energy exchange system that incorporates spatial configuration. An outline of the model framework and system boundary is shown in Fig. 1. Here, we focus on spatial problems, such as energy decay for steam extraction and transportation, and new infrastructure construction costs. For the assessment of IS systems, there are three key issues:

- New infrastructure investment costs should be included, particularly those of pipelines and related equipments for transmitting heat energy;
- Energy conversion efficiency is expected to improve the utilisation of exhaust heat from thermal power plants. However, extracting steam from the Rankine cycle of the nuclear reactor inevitably reduces the amount of heat that is converted into electricity. Heat energy properties should also be considered, such as heat loss during pipeline transportation or the power used for conveyance;
- The heat energy demand fluctuates on a seasonal or daily basis depending on customers' needs. In our case, daily fluctuations are predictable since the heat would be used for continuous processes. Seasonal fluctuations will however affect the diameter of the pipeline needed. The energy simulation has thus been carried out based on the aggregated energy demand per year, taking into consideration the working schedules of factories (seasonal and weekly fluctuations, enabling the equivalent-full time utilisation of industrial processes to be calculated).

The model system consists of three sub-models: an energy consumption system, an energy generation system and an energy distribution system. The energy consumption system sub-model consists of the factories consuming the steam. The energy generation system sub-model represents the extraction of steam from the Rankine cycle of the nuclear reactor. In the energy distribution system sub-model, equivalent-CO_2 emissions and energy transfer costs are calculated taking heat losses into consideration. These models were formulated as a linear programming problem using Python.

Energy consumption system

This system is mainly the result of the calculations described in Section 2.2. Q_{IS}, the maximum thermal power required to cover the annual heat needs of all consumers (MW_{th}), is expressed in formula (4):

$$Q_{IS} = \sum \frac{C_{i,j}\ 10^3}{NH_{FT,j}} \tag{4}$$

Where $C_{i,j}$ is the steam consumed by factory i in subsector j ($GWh_{th}/year$) and $NH_{FT,j}$; and $NH_{FT,j}$ is the average number of hours per year during which facilities in subsector j are operated.

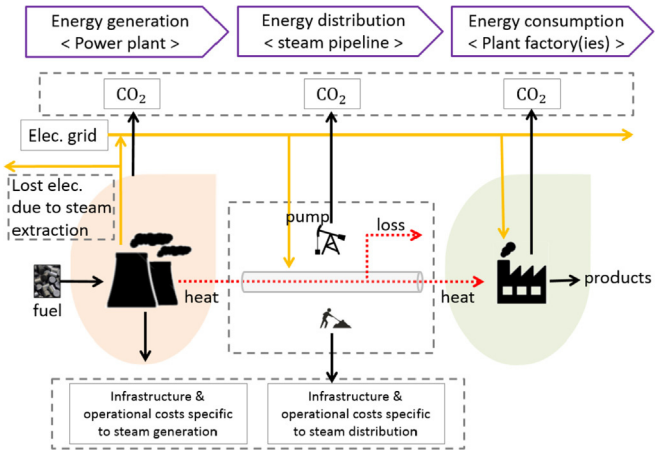


Fig. 1. Structure of the regional energy analytical model and system boundaries. Notes: The parameters assessed in this paper are shown in boxes with dotted lines.

Energy generation model

The cost of steam generation with the nuclear plant is regarded as the investment cost of a Generation III PWR attributable to steam generation. The infrastructure cost, C_{CU} (€/MW_{th}), of a combined heat and power (CHP) upgrade in a PWR is expressed as in formula (5):

$$C_{CU} = Q_{IS} C_{MCU} \tag{5}$$

Where Q_{IS} is the maximum thermal power required (MW_{th}); C_{MCU} is the marginal cost of a CHP upgrade in a PWR (€/MW_{th}). The Energy Technology Institute (ETI, 2016) states that C_{MCU} is between 0.05 €/MW_{th} and 0.09 €/MW_{th}, including e.g. pumps and condensers. As a conservative assumption, it is assumed here that C_{MCU} is 0.1 €/MW_{th}. Operational and maintenance costs are assumed to represent 1% of the capital cost.

The model takes into account the reduction in power generation due to heat extraction from the Rankine cycle of a PWR. For the extraction of steam at 285 °C and 70 bar at the steam generator outlet, E_{LS}, the electricity loss (MWh_e) represents one third of the thermal power generated (IAEA , 2016). This represents the opportunity cost of operating a nuclear plant in a cogeneration mode. In this paper, we considered the most conservative case in which the heat loads always occur when the plant is providing electricity to the power grid. This is not true in practice since the load factor of nuclear plants is rarely of 100%, the French average for 2017 being 72% (CEA (French Nuclear and Alternative Energies Commission), 2017). A more detailed analysis, based on a real, experimental project, would want to consider the temporal correlation between electricity and heat production. In the frame of this study however such precise, case by case data were not available. Given the uncertainty at stake, retaining the conservative assumption mentioned above is reasonable.

Given the above assumptions, the required cost (C_{pp})and additional CO_2 emissions (E_{pp}) to compensate for power generation losses are expressed as equations (6) and (7) respectively:

$$C_{pp} = C_{el} E_{LS} \tag{6}$$

$$E_{pp} = e_{el}\, E_{LS} \tag{7}$$

where C_{el} is the electricity generation cost in a nuclear power plant (€/MWh_e), and e_{el} is the equivalent-CO_2 emission factor of the electricity (t eCO_2/GWh_e; see Table 2).

Energy distribution model

The energy distribution model assesses costs and equivalent-CO_2 emissions as well as the infrastructure construction cost required for energy transfers. The 250 °C steam extracted from the power plant would be transported to heat sinks through a single superheated pipe, the most commonly used type of pipeline for 250 °C steam transportation (García-Gutiérrez et al., 2015; Wang et al., 2017). Heat energy flows out from the pipeline surface during transport, and this loss should be assessed under various conditions (e.g. pipe diameter, insulation thickness, distance). Furthermore, electricity is needed for pumping, to circulate the heating medium.

a) Pipe diameter

The pipe diameter is calculated using empirical data on existing pipeline networks transporting steam at 250 °C (García-Gutiérrez et al., 2015; Wang et al., 2017). First, the steam flow G_s (kg/s) and the volumetric flow rate $\mathring{v}$ (m^3/s) are calculated as in equations (8) and (9):

$$G_s = \frac{Q_{IS}}{\lambda} 10^6 \tag{8}$$

$$\mathring{v} = \frac{G_s}{\rho} \tag{9}$$

Where Q_{IS} is the maximum thermal power required (MW_{th}); λ is the latent vaporisation heat (Ws/kg); ρ is the water density (kg/m^3). Here, ρ is assumed to be 6.5 kg/m^3, as in e.g. Wang et al. (2017).

Then, the pipe diameter D (m) is calculated using formula (10):

$$D = \sqrt{\left(\frac{4\,\mathring{v}}{\pi\, v}\right)} \tag{10}$$

where v_{max} is the flow velocity (m/s). If the steam flows at a low velocity or stagnates, there could be serious accidents (e.g. condensation induced water hammer). 21 m/s is a reasonable assumption to ensure safety of operation (García-Gutiérrez et al., 2015; Wang et al., 2017).

b) Heat loss assessment

To calculate heat loss, the pipeline energy balance must be understood. Referring to Yoshiyuki et al. (2001), the heat balance at a microscopic interval, dx, is expressed as equation (11). Integrated along pipe length l, equation (12) describes TH_{LS}, the thermal loss from heat transfer(W_{th}):

$$cG_s \frac{dT}{dx} = \pi\, DK(T_{out} - T) \tag{11}$$

$$TH_{LS} = cG_s(T_s - T_d) = cG_s(T_s - T_{out})\left\{1 - e^{-\left(\frac{\pi\, DK}{cG_s}\right)l}\right\} \tag{12}$$

Where T_s is the steam temperature at the supply-side; T_d is the steam temperature at the demand side; T_{out} is the outside temperature; c is the steam heat ratio (W/(kg.K)); G_s is the steam flow (kg/s); D is the pipeline diameter (m); l is the length of the pipeline (m); and K is the overall heat transfer coefficient (W/m^2.K). According to García-Gutiérrez et al. (2015) and Wang et al. (2017), a well-insulated pipeline can have a low heat transfer coefficient (0.24 W/m^2.K), but most pipelines used in practice, especially long-distance overheated pipelines, have high heat transfer coefficients (1.2–1.5 W/m^2.K). Here, K is assumed to be 0.5 W/m^2K.

c) Pumping cost and CO_2 assessment

The operation of a pumping system is required for transferring steam. The energy consumption of the system (W_e), P_{PM}, is expressed as in equation (13):

$$P_{PM} = \frac{g\, G_s H}{\eta_p} \tag{13}$$

Where g is the gravitational acceleration (m/s^2), η_p is the pump efficiency ratio (0.75), G_s is the steam flow (kg/s), and H is the lifting height (m). Referring to the Darcy–Weisbach equation, the lifting height is expressed as equation (14):

$$H = f\frac{L}{D}\frac{v^2}{2g} + c \tag{14}$$

Where L is the pipeline length (m), D is the pipeline internal diameter (m), v is the velocity of the medium (m/s), g is the gravitational acceleration (m/s^2), f is the friction loss coefficient, and c is

the friction loss by local resistance (0.5).

From the above assumptions, and assuming that the system is operational 100% of the year (8760 h/year), the required costC_{PM}, and additional equivalent-CO_2 emissions, E_{PM}, involved in the operation of a pumping system are expressed in equations (15) and (16) respectively:

$$C_{PM} = C_{el} P_{PM}\ 8760 \tag{15}$$

$$E_{PM} = E_{el}\ P_{PM}\ 8760 \tag{16}$$

where $C_{el}$is the generation cost in a nuclear plant (€/MWh_e), and E_{el} is the equivalent-CO_2 emission factor (t eCO_2/GWh_e).

In practice, nuclear plants would not provide heat throughout 100% of the year. The equivalent full-time utilisation would depend on the heat demand profile of the supplied factories (see Table 3) and plant outages schedules. Assuming the conservative value of 100% (slightly) increases the costs and CO_2 emissions from pumping, which is reasonable considering the uncertainty at stake.

d) Pipeline installation cost

Dalla Rosa et al. (2012) and Hirsch et al. (2016) provide an equation expressing the pipeline cost as a function of the inner diameter. It represents the sum of the pipe, insulation layer, pumping stations, civil engineering work, sand filling and labour costs needed to build a pipeline designed to transport hot water (80–120 °C). Discussions with French district heating companies highlighted that the infrastructure cost should be 2 to 3 times higher for the construction of pre-insulated industrial pipes designed to carry superheated steam (250 °C), and the conservative value of 3 is used in this paper. Given these considerations, C_{SP}, the capital cost of the single superheated pipeline (€/m) is expressed in equation (17), where D is the pipe diameter (m):

$$C_{SP} = 4500\, D^2 + 6000\, D + 900 \tag{17}$$

The operational and maintenance costs are assumed to represent 4% of the total pipeline cost (Dalla Rosa et al., 2012).

Infrastructure costs related to steam generation and steam transport sub-models were converted to annual rental cost r_{IF} (€/MWh_{th}) applying a discount rate method, expressed as equation (18):

$$r_{IF} = I_{CC} \frac{i}{1 - \left(\frac{1}{1+i}\right)^t} \tag{18}$$

Where I_{CC} is the infrastructure capital cost (€/MWh_{th}), either for the CHP upgrade of a PWR (C_{CU}, see equation (5)) or for the superheated steam pipeline (C_{SP}, see equation (17)). i is the interest rate and t is the time (year).

3. Results and discussions

Each sub-section below (3.1, 3.2 and 3.3) present and discuss the results based on the methodological steps presented in the earlier sub-sections (2.1, 2.2 and 2.3, respectively). Sub-sections 3.1, 3.2 and 3.3 together answer the three research questions raised in sub-section 1.3.

3.1. Relevant industrial subsectors for using 250 °C power plant-sourced steam

Based on the data and assumptions presented in Section 2.1, Table 3 shows the parameters characterising the subsectors that could constitute relevant targets for those low carbon policies which aim at encouraging the external supply of 250 °C steam to factories. The application of the restrictive criterion shown in 2.1 leads to just 14 of the 114 subsectors of the French industrial sector being selected for further analysis. The greatest potential for the creation of IS complexes is in subsectors in which the following parameters are relatively high:

- The share of heat consumption at temperatures lower than 250 °C (which indicates the potential of external supply of steam at 250 °C to replace all or part of the on-site steam generation);
- The average equivalent-CO_2 emissions from heat generation with on-site boilers (which determine the potential for reducing equivalent-CO_2 emissions by replacing on-site steam generation);
- The average consumption of heat <250 °C per factory (which affects the cost of the external steam supply);
- The share of heat costs in the production value (which affects the interest of factory owners in reducing the cost of steam).

Table 3
Industrial subsectors suitable for the integration of IS complexes.

Subsector	NACE code	Average consumption of heat <250 °C per factory ($GWh_{th}/year$)	Equivalent full-time utilisation (% of year)	Cost of heat generation < 250 °C over production value (see equation (1))	Share of heat <250 °C in total heat consumption	Equivalent-CO_2 emissions ($t\ eCO_2/GWh_{th}$)
Fruit & vegetables	1031Z-1039A-B	19	66%	1.1%	93%	548
Dairy products	1051A-B-C-D	32	74%	1%	94%	564
Starch products	1062Z	704	66%	7.3%	76%	549
Sugar refinery	1081Z	97	34%	6.6%	82%	599
Malt production	1106Z	88	66%	3.2%	100%	522
Wood panels	1621Z	39	65%	3.9%	71%	554
Pulp & paper	1711Z	695	87%	36.9%	91%	595
Card & paper	1712Z	133	87%	8.8%	87%	589
Corrugated card	1721A-B	15	87%	1.3%	99%	596
Dyes & pigments	2012Z	62	87%	3.7%	76%	587
Other basic organic chemicals	2014Z	136	87%	2.8%	57%	591
Basic pharmaceutical products	2110Z	27	87%	1.0%	92%	570
Basic plastic materials	2016Z	41	82%	1.1%	78%	566
Synthetic rubber	2017Z	131	82%	2.2%	54%	588

Notes:Highlighted in green: Agri-food industry; Highlighted in red: Wood, pulp and paper; Highlighted in blue: Chemical and pharmaceutical industries; Highlighted in yellow: Plastic, rubber and other elastomers.

3.2. Spatial and sectoral distribution of relevant factories

Fig. 2 shows the spatial mapping of nuclear sites and factories in the subsectors identified in Table 3. Using the calculations described in equations (2) and (3), the amount of heat consumed annually at temperatures below 250 °C has been assessed for each industrial facility. Approximately 30 TWh_{th}/year of heat below 250 °C is consumed in factories located less than 100 km from a nuclear site. Given the CO_2 emissions of each subsector, it has been estimated that the generation of heat below 250 °C within these factories is responsible for the emission of 16,000 kt eCO_2/year. This is equivalent to 20% of the total equivalent-CO_2 emissions from the French industrial sector and 3.5% of the total French emissions (and 25% of emissions from the industrial sector; Ministère de l'environnement, de l'énergie et de la mer, 2017).

Figs. 3 and 4 show respectively the sectoral and spatial distribution of the heat below 250 °C consumed by the factories in the subsectors studied. Regions with an aggregated industrial heat consumption of less than 100 $GW_{th}/year$ have been discarded and are not shown in Fig. 4. These represent less than 1% of the total. Figs. 2–4 constitute a solid basis for analysing the regional potential of IS complexes using steam sourced from nuclear plants in France. When looking at these Figures, one should however keep in mind that they do not allow to distinguish those IS complexes which are economically feasible from the others. Figs. 2–4 considers the heat demand of all factories located in a 100 km radius from nuclear sites, while in practice transporting the steam over distances exceeding 3–5 km may not be cost-effective (NISP , 2008). Compared to the overall regional potential assessed in Figs. 2–4, the potential for cost-effective implementation of IS complexes based on nuclear plants is limited given the location of existing plants. Under the current spatial configuration, relocation of the factory may be of interest for the card & paper and the malt factories located 3.2 km and 2.4 km from *Nogent-Sur-Seine*, respectively. The two chemical plants situated 1.8 km away from *Le Bugey* constitute another promising IS complex. The *Gravelines* area, with a pharmaceutical plant located 0.5 km away from the thermal plant, is also relevant.

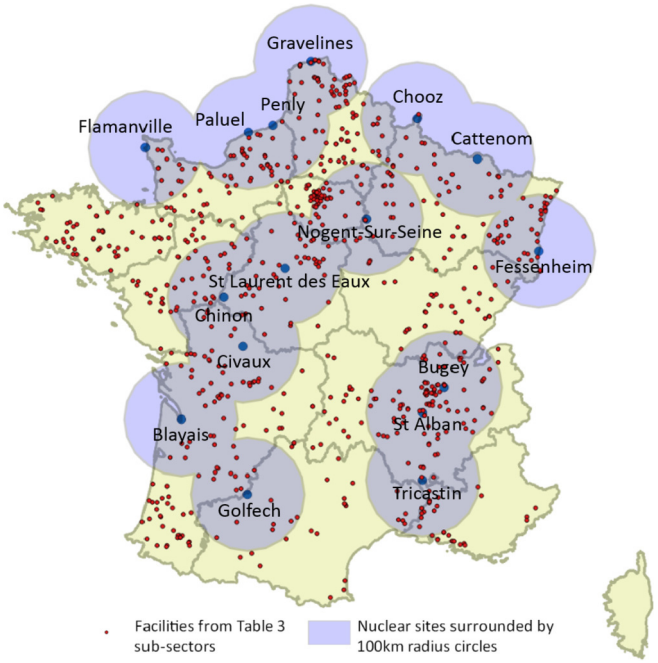


Fig. 2. Spatial mapping of nuclear sites and factories studied.

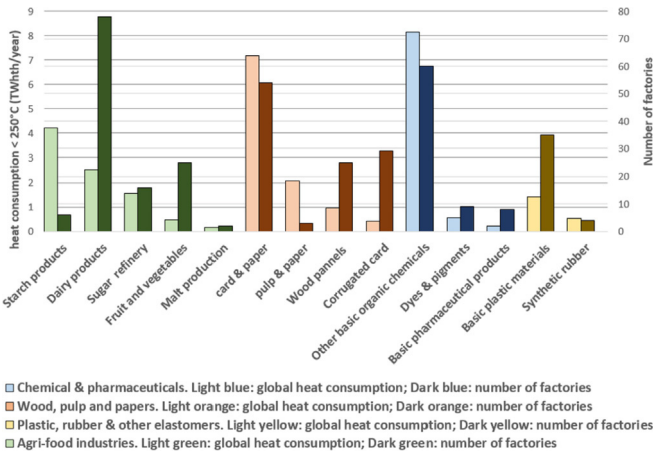


Fig. 3. Subsectoral distribution of heat consumption below 250 °C and of corresponding factories, for the land areas shown in Fig. 2. Notes: The average heat consumption of a single factory is shown in column 3 of Table 3.

To increase the number of feasible IS systems, ambitious urban planning policies would be needed at a regional level, so as to relocate the factories from relevant sub-sectors (see Table 3) in contingent areas centered on nuclear plants. The more contiguous the distribution of various factories within the same geographical area, the more likely IS are to emerge (Jensen, 2016). Those policies which aim at relocating factories from different sub-sectors within a contiguous area can thus expect positive spillover effects such as the sharing of services or additional recovery of waste materials (see e.g. Chertow, 2000).

3.3. Techno-economic assessment of industrial symbiosis considering spatial configuration

Sub-Section 3.3 aims to determine the conditions upon which IS systems based on nuclear plants may become economically attractive compare to the usual heating solutions. To analyse and discuss the techno-economic feasibility of thermal plant-sourced steam transfer to nearby factories, two theoretical cases are considered, as described in Table 4. IS complex 2 is not feasible in practice given the current location of the factories (located too far away from the thermal plant, see Table 4). It is however interesting to evaluate the cost and climate savings that could exist if the factories identified in IS complex 2 would be relocated in the direct proximity of the *Nogent-Sur-Seine* nuclear site. As shown by Togawa et al. (2014), reducing energy costs may represent a sufficient incentive for a factory to be located near a thermal power plant, especially in a context of increasing pressure to reduce the environmental impact and dependence on fossil fuels.

According to equations (7) and (16), the equivalent-CO_2 emissions avoided by the implementation of IS complexes 1 and 2 would be approximately 78 kt eCO_2/year and 377 kt eCO_2/year respectively. The total amount of emissions avoided depends on the distance over which the steam is transported (as the pumping power increases with the distance), but the effect is relatively small when replacing fossil fuels with power plant-sourced steam. The cost of the steam supplied was calculated based on a reference generation cost of electricity from French nuclear plant of 42 €/MWh_e(Ministère de l'environnement, de l'énergie et de la mer, 2016). Assuming a discount rate of 3.5% over 20 years, as recommended by the EC (2014b), and according to equation (18), the results are shown in Fig. 5 and compared to the business as usual (BAU) case where steam at 250 °C is generated using conventional

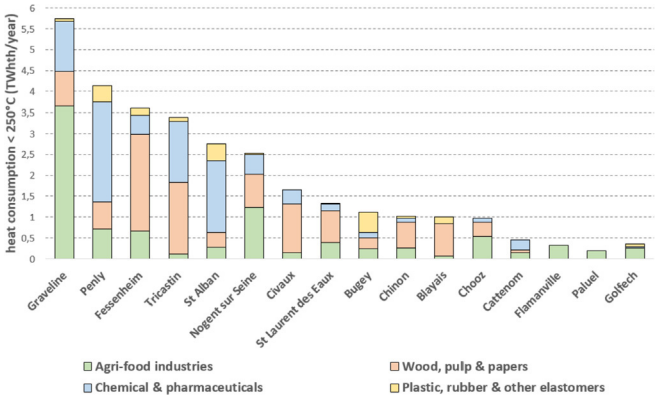


Fig. 4. Spatial distribution of heat consumption below 250 °C, for the land areas shown in Fig. 2. Notes: The average heat consumption of a single factory is shown in column 3 of Table 3.

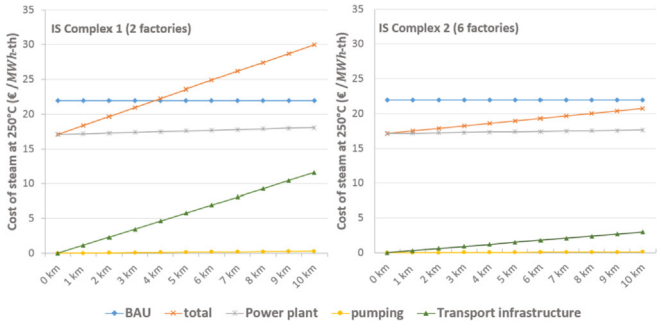


Fig. 5. Relationships between distance and cost of steam in IS complexes 1 and 2.

individual heating systems. In this paper, the reference is the average cost paid by large industrial users for the generation of steam at 250 °C, 22 €/MWh$_{th}$ (INSEE, 2014). In practice, this cost ranges from 12 €/MWh$_{th}$ to 35 €/MWh$_{th}$ depending on parameters specific to each factory (e.g. the region, the workforce and the subsector; see INSEE, 2014).

Fig. 5 shows that the distances over which the steam is transported should not exceed 4 km to ensure cost-effectiveness of the steam supply in IS complex 1 (transportation heat losses represent 1.1% and 2.3% of the heat generated, respectively for 1.8 km and 4 km distances; following equations (11) and (12)). The cost of the steam in IS complex 2 is less sensitive to the distance than in IS complex 1, and may be competitive if the length of the pipeline required to connect the 6 factories to the thermal plant does not exceed 10–11 km (heat losses equal 2.7–2.9% of the heat generated; such low heat losses however require well insulated pipelines, as discussed in Section 2.3). The large amount of steam supplied in IS complex 2 (five times higher than in IS complex 1; see Table 4) may justify the transport of heat over long distances. This highlights the importance of coordination and planning policies, which should encourage the location of relevant factories within contiguous areas.

Because the electricity losses are approximately one third of the steam extracted at the steam generator outlet (see equation (6)), the main part of the cost and equivalent-CO_2 emissions occurs within the power plant boundaries. This means that the cost of the steam is highly dependent on the cost of electricity generation in the power plant, which is expected to be higher than 42 €/MWh_e for future nuclear plants (CRE (French Energy Regulatory Commission), 2013). However, fossil fuel prices are also expected to rise (see e.g. IEA (International Energy Agency), 2016); and the French public authorities are willing to establish a carbon price of 100 €/t eCO_2 by 2030. It is also possible to extract 120 °C heat from the secondary turbine of the power plant, thus reducing power loss to one sixth of the thermal output. This would increase the cost-effectiveness of heat supply, but reduce the range of industrial applications, so that the potential for decarbonising the industrial sector would be reduced. Future research should examine these market and techno-economic issues in more detail. At this temperature level (100–120 °C), another plausible use of the heat from nuclear plants is space heating and domestic hot water supply to residential and commercial buildings (through district heating networks; see e.g. Leurent et al., 2018a, 2018b; Jasserand and Lavergne, 2016). When considering such applications, it must be kept in mind that other low carbon, potentially cost-effective heating solutions do exist (e.g. ground source heat pumps; see Esen et al., 2007, 2006; or Esen and Yuksel, 2013), and these must be included in the comparison.

It was calculated that implementing a steam exchange system between the two plants of IS complex 1 and *Le Bugey* thermal plant could generate about 450 k€/year of energy savings (reducing annual heat costs by 15%), with a payback period of 8 years for recovery of the infrastructure investment costs (the costs of relocating the plants are not including). Small IS systems (e.g. IS complex 1, or connecting the *Gravelines* to the pharmaceutical factory located 0.5 km away) could serve to demonstrate the benefits of such complexes. Small-scale projects are more likely to succeed than large projects involving many stakeholders, which can be penalized by the diversity of conflicting interests or the diversity of technical requirements (see e.g. Tudor et al., 2007). The power plant could serve as the key organisation around which discussions are organised. Government subsidies for pipeline construction would also be an important incentive for the implementation of large steam networks (as emphasised by e.g. Togawa et al., 2014; Tudor et al., 2007). A broad sense that the project is supported by national and local authorities would also be required.

Once a first IS system is implemented, and if it proves to be overall good to the society, policy makers and stakeholders could build on the experience to promote more ambitious IS projects implying the relocation of plant factories closer to nuclear sites.

Table 4
Theoretical IS complexes assessed.

	IS Complex 1	IS Complex 2
Number of factories	2	5
Products manufactured by the factories (and distance from the closest nuclear site)	Basic organic chemicals (1.8 km)	Card & Paper (3 km), Malt (3 km), card & paper (35 km), pulp & paper (60 km), basic organic chemicals (69 km)
Name of the closest nuclear site	*Le Bugey*	*Nogent-Sur-Seine*
Total amount of heat < 250 °C consumed ($GW_{th}/year$)	137	661
Equivalent-CO_2 emissions from heat generation < 250 °C using conventional individual heat systems ($kt\ eCO_2/year$)	82	696

Future research could aim to analyse how a niche for the implementation of a demonstration project could be created and then developed to catalyse a generalization of IS complexes. This would require the identification of the actors and institutions who could facilitate learning processes, establish shared visions of the future and form supportive networks.

4. Conclusions

Based on a detailed spatial analysis, this paper assesses the technical and economic feasibility of transferring steam sourced from nuclear plants to factories in France. The results are relevant for policies exploring the relationships between energy planning and urban design. Integrated land use and regional energy planning needs to be taken into consideration during the design process. Three points are highlighted:

- 14 industrial subsectors are identified as relevant for using power plant-sourced steam at 250 °C. The involvement of large, long-term heat consumers such as manufacturers of starch products or pulp and paper would support the economic viability of IS complexes in the long term.
- Within a 100 km radius of existing nuclear reactors, factories in these 14 subsectors use 30 TWh_{th}/year of heat below 250 °C. On-site heat generation, currently based on fossil fuels, is responsible for approximately 3.5% of French equivalent-CO_2 emissions.
- Power plant-sourced steam could reduce energy supply costs and equivalent-CO_2 emissions in locations close to these thermal power plants. As a concrete example, it was shown that implementing a steam exchange system between two chemical plants and *Le Bugey* thermal plant, 1.8 km away, could reduce the factories annual heat costs by approximately 15%. Expected cost reductions could affect the location of new factories, especially energy-intensive industries.

This study has been conducted using empirical data and planning examples. Future research could address a number of additional technical, economic and environmental issues:

- As this study focused on a broad techno-economic analysis, the results obtained cannot be directly applied to real cases. For implementation in real planning, some technical hypotheses should be assessed experimentally first (such as performed by e.g. Esen et al., 2007, 2006; or Esen and Yuksel, 2013, in the case of heat supply to individual houses). Heat from sources must be transported to users in pipes economically and efficiently. This is not an easy task, especially in a pipe network connected to multiple sources and multiple users;
- A different approach is required to determine the market opportunities and challenges inherent to each of the subsectors studied, so that a long-term IS strategy can be planned. Constructing a network and engaging stakeholders such as energy companies, factory owners, local municipalities and so on, is an important consideration (Hein et al., 2017). In this study we assessed the overall cost and equivalent-CO_2 emissions, rather than individual stakeholder profit or costs. Such an analysis is necessary for future research. Nonetheless, the data collected for this study do represent a significant added value, facilitating future research on IS complexes.
- Given the specific nature of the IS studied (based on nuclear plants), a broader assessment of how such IS would affect the environment is required. It is recognised that, by establishing several hydraulic barriers in the form of heat exchangers across which heat is transferred from plants to factories, the risk of hazardous exposure of humans to radioactivity via the transportation system, due to leakage, can be reduced to almost zero (STUK, 2009). Thorough engineering analysis would be required, ensuring compliance with the safety principles stated by Fortum (2013). Issues related to public opinion should also be considered as well as the attitude of the French nuclear safety regulator toward the localisation of industries near nuclear power plants.
- This paper only considers the cost of 'cogeneration readiness' in the nuclear plants. 'Cogeneration readiness' is a term coined by ETI (2016) to name the equipment required in a nuclear plants to enable heat supply (e.g. heat exchangers, pumps). This paper accounts for the capital costs specific to heat (generation and transportation), but excludes the costs attributable to electricity production. Therefore, it does not allow us to reach any conclusion as to whether nuclear plant-sourced steam transfer to industrial sinks offers more benefits than other systems generating the same amount of electricity and heat. To answer this question, further analysis would be needed taking into account the entire investment cost of a new nuclear power plant. A reasonable conclusion is that, if new nuclear plants are planned, energy systems should be optimised so as to make the most efficient use of the produced energy.

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